

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCM1363 DISCRETE MATHEMATICS

FRIDAY, 23 SEPTEMBER 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates :

Answer ALL questions in **section A**.

Answer ANY ONE question in **section B**.

If more than ONE question from **section B** is answered, then only the **first question attempted in section B will be marked**.

All questions carry equal marks. Marks allocated for each part of the questions are indicated in brackets.

UCCM1363 DISCRETE MATHEMATICS**SECTION A**Answer ALL questions.

- Q1. (a) Use a truth table to determine the validity of the following arguments.

If the road goes to the capital, then the bus stop is not here.
 If the bus stop is here, then the road does not go the capital.
 Therefore, the road goes to the capital, or the bus stop is here.

(10 marks)

- (b) For each of the following statement, rewrite the statement informally without using variables or the symbols $\forall$ or $\exists$. Then indicate whether the statement is true or false with explanation.

(i) $\forall$ integers x , $\exists$ an integer y such that $x + y = 0$. (5 marks)

(ii) $\exists$ an integer x such that $\forall$ integers y , $x + y = 0$. (5 marks)

- (c) Without using a truth table, show that $p \vee \sim q \Rightarrow r \vee q \equiv (\sim p \wedge q) \vee (r \vee q)$. (5 marks)

[Total : 25 marks]

- Q2. (a) Suppose x and y are any integers with $x \neq y$. Use mathematical induction to prove that $(x^n - y^n)$ is divisible by $(x - y)$ for $n \geq 0$. (8 marks)

- (b) Suppose a certain amount of money is deposited into an account paying 10 % annual interest, compounded quarterly. For each positive integer n , let S_n be the amount on deposit at the end of the n th quarter, and let S_0 be the initial amount deposited. If $S_0 = \text{RM } 5000$, find the amount of money on deposit at the end of 4 years. (7 marks)

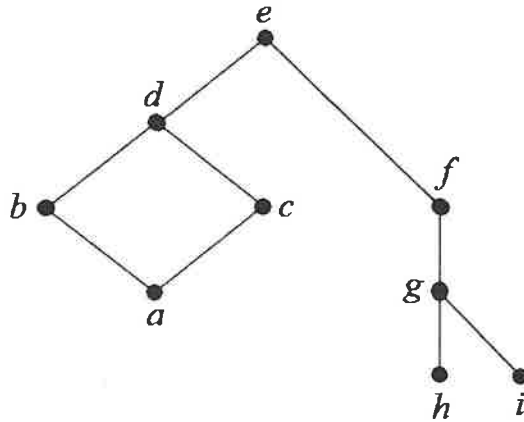
- (c) Use iteration to guess an explicit formula for the following sequence.
 $a_n = 2a_{n-1} + 3$, for all integers $n \geq 1$, $a_1 = 2$. (10 marks)

[Total : 25 marks]

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- Q3. (a) Let A = the set of real numbers and consider the following relation R on A : xRy if and only if $4x^2 + y^2 = 36$. Find $R(\{3, 4, 5, 6\})$. (6 marks)

- (b) Consider the following poset:



Determine the maximal, minimal, greatest, and least elements. (6 marks)

- (c) (i) Determine the Hasse diagram of the relation on $A = \{4, 16, 64, 256, 1024\}$ whose matrix is:

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

(5 marks)

- (ii) Identify whether this Hasse diagram is a lattice and explain your answer. (2 marks)

- (d) Explain how to determine whether a lattice is:

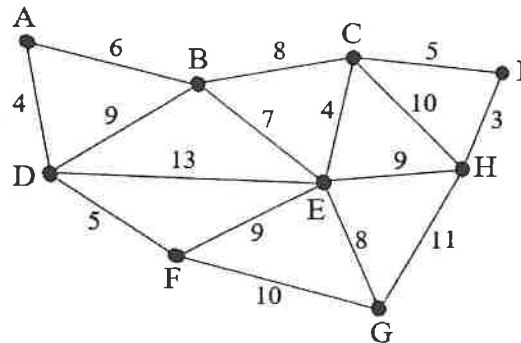
(i) distributive. (3 marks)

(ii) complemented. (3 marks)

[Total : 25 marks]

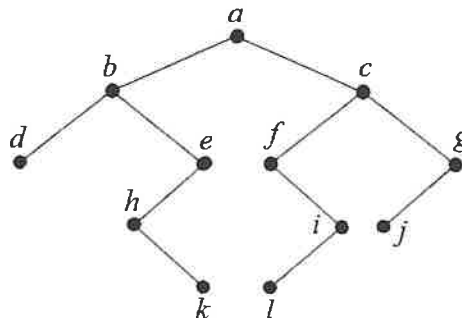
UCCM1363 DISCRETE MATHEMATICS**SECTION B (Answer only ONE question)**

Q4. (a) Find a minimal spanning tree for the following graph by using:



- (i) Prim's algorithm, where A is chosen as a starting point. (9 marks)
- (ii) Kruskal's algorithm. (9 marks)
- (b) Give one difference between Prim's algorithm and Kruskal's algorithm. If point E is chosen as the starting point in Prim's algorithm, will the weight calculated in (a)(i) remains the same? (3 marks)
- (c) Give four applications of minimal spanning tree in real life. (4 marks)
- [Total : 25 marks]

- Q5. (a) Discuss the specified orders followed in the process of visiting each vertex of a tree in preorder search, inorder search and postorder search. (3 marks)
- (b) Perform preorder search, inorder search and postorder search of the tree below:



(18 marks)

- (c) Give four applications of binary search tree. (4 marks)
- [Total : 25 marks]

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Formula Sheet

Let p and q be statement variables:		Rules of Inference	Name
(1) The negation of $p \Rightarrow q$ is $p \wedge \sim q$.		1. $p \Rightarrow q$ p _____	Modus Ponens (Law of Detachment)
(2) The contrapositive of $p \Rightarrow q$ is $\sim q \Rightarrow \sim p$.		$\therefore q$	
(3) The converse of $p \Rightarrow q$ is $q \Rightarrow p$.		2. $p \Rightarrow q$ $\sim q$ _____	Modus Tollens
(4) The inverse of $p \Rightarrow q$ is $\sim p \Rightarrow \sim q$.		$\therefore \sim p$	
Theorem 1.1 Let p and q be statements variables. (1) $p \Rightarrow q \equiv \sim p \vee q$ (2) $p \Rightarrow q \equiv \sim q \Rightarrow \sim p$ (3) $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$		3. p or q _____	Disjunctive Addition
Theorem 1.2 (1) Commutative laws: $p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$		$\therefore p \vee q$ or $\therefore p \vee q$	
(2) Associative laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$		4. $p \wedge q$ or $p \wedge q$ _____	Conjunctive Simplification
(3) Distributive laws: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$		$\therefore p$ or $\therefore q$	
(4) Identity laws: $p \wedge t \equiv p$ $p \vee c \equiv p$		5. $p \vee q$ or $p \vee q$ $\sim p$ or $\sim q$ _____	Disjunctive Syllogism
(5) Negation laws: $p \vee \sim p \equiv t$ $p \wedge \sim p \equiv c$		$\therefore q$ or $\therefore p$	
(6) Double negative laws: $\sim(\sim p) \equiv p$		6. p q _____	Conjunction
(7) Idempotent laws: $p \wedge p \equiv p$ $p \vee p \equiv p$		$\therefore p \wedge q$	
(8) Universal bound laws: $p \vee t \equiv t$ $p \wedge c \equiv c$		7. $p \Rightarrow q$ $q \Rightarrow r$ _____	Hypothetical Syllogism
(9) De Morgan's laws: $\sim(p \wedge q) \equiv \sim p \vee \sim q$ $\sim(p \vee q) \equiv \sim p \wedge \sim q$		$\therefore p \Rightarrow r$	
(10) Absorption laws: $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$		8. $p \vee q$ $\sim p \vee r$ _____	Resolution
(11) Negations of t and c : $\sim t \equiv c$ $\sim c \equiv t$		$\therefore q \vee r$	

